

Concentrated output sensitivity in connectome-scale biological models

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Abstract

Mechanistic models can reproduce biological outputs while leaving their underlying parameters uncertain. We quantify how sensitivity concentrates across parameter combinations using complete spectra, matched comparisons of measured outputs and model states, and fitting experiments with known generating parameters. In a nematode circuit with 3076 connection weights, 26 combinations carry 90% of motor-voltage sensitivity and 66 carry 99%. The concentration estimate changes with output reduction and perturbation scale in a walking controller, and across outputs derived from the same nematode connection perturbations. In a matched fly visual-network comparison, sensitivity is more concentrated in the trained state than in the initialized state. In a separate gain assay, combining stimulation conditions records additional sensitivity directions while the spectrum remains strongly concentrated. In cell models with known generating parameters, a concentrated sensitivity spectrum accompanies both accurate recovery and output-matched fits with substantial parameter error. A prospective simulated stimulus comparison did not demonstrate an average prediction benefit from the tested ensemble-selection rule. Sensitivity concentration alone can miss directions added by a new assay and cannot specify achieved parameter accuracy.

Introduction

Mechanistic models connect biological components to the outputs they collectively generate. A neural circuit model maps synaptic weights and membrane conductances to activity; a biochemical pathway maps rate constants to a signalling response. Fitting these outputs makes a model useful for reproducing observations and testing perturbations. It also raises a question: what does the measured output reveal about the model’s internal parameters?

Different parameter sets can produce similar dynamics. Conductance-based neural models provide well-studied examples of this degeneracy (1–4). Sloppy-model theory describes a related feature: output sensitivity can differ greatly between parameter combinations, so that a few strongly sensitive combinations dominate a much larger space (5–7). This concentration is distinct from identifiability, which concerns whether observations determine parameter values under a specified model and inference protocol (8–11). Large pathway models can retain useful predictive accuracy despite substantial parameter uncertainty (12, 13).

Connectome-constrained models now make it possible to examine this distinction at the scale of hundreds of neurons and thousands of connection weights. Wiring diagrams (14–16), biophysical simulation frameworks (17–19), and measured kinematics (20, 21) support models of nematode movement and fly sensorimotor function (22–25). Their outputs can be perturbed repeatedly while controlling the parameters, input and readout. Neural activity, motor commands and body movement nevertheless remain different measurements of the same underlying system. These models

provide a setting to examine whether changing an assay reveals additional sensitivity directions, changes their concentration or improves parameter recovery.

Here, we quantify concentration from the complete connection-weight spectrum of a nematode circuit and compare outputs derived from the same perturbations. Matched measurements of a walking controller and a fly visual network test how readout and model state change that concentration. By combining stimulation conditions, we test whether added sensitivity directions are captured by the concentration count. Known-truth recovery experiments then examine whether a similar count implies similar parameter accuracy. A prospective stimulus comparison tests whether choosing an additional observation from a baseline-fitted model ensemble improves prediction.

Results

Output sensitivity at connectome scale

We measure the change in a specified output vector when each parameter is perturbed in turn. The resulting finite-difference Jacobian J records how parameter changes map onto output changes. A fixed diagonal weighting W standardizes those changes, and the Gram matrix $H = J^\top W J$ summarizes squared sensitivity along parameter combinations. The counts d_{90} and d_{99} are the smallest numbers of leading combinations carrying 90% and 99% of the total squared sensitivity. We call these concentration counts. Numerically resolved directions have singular values above a stated numerical tolerance.

The primary assay uses the BAAIWorm point-neuron circuit, with 136 cells and 3076 chemical-synapse and gap-junction weights. A fixed sinusoidal sensory drive supplies the input, and 80 motor-neuron voltages sampled over time supply 9600 output components. At relative perturbation step 0.5, $d_{90} = 26$ and $d_{99} = 66$, with participation ratio 6.83, a continuous measure of the effective number of contributing sensitivity directions (Fig. 1b). Thus, fewer than 1% of parameter combinations carry 90% of the squared motor-voltage response. Perturbation-step controls distinguish strong sampled responses from an unresolved weak response (Supplementary §2).

Chemical synapses contribute 69% of squared motor-voltage sensitivity from 1992 weights, compared with 31% from 1084 gap-junction weights. The mean squared sensitivity per chemical connection is 1.22 times the gap-junction value, while the median column norm, measuring response magnitude per connection, is 6.62 times greater. The difference between these ratios shows that typical connection sensitivity and total class contribution describe different aspects of the distribution.

a Measurements, sensitivity and recovery

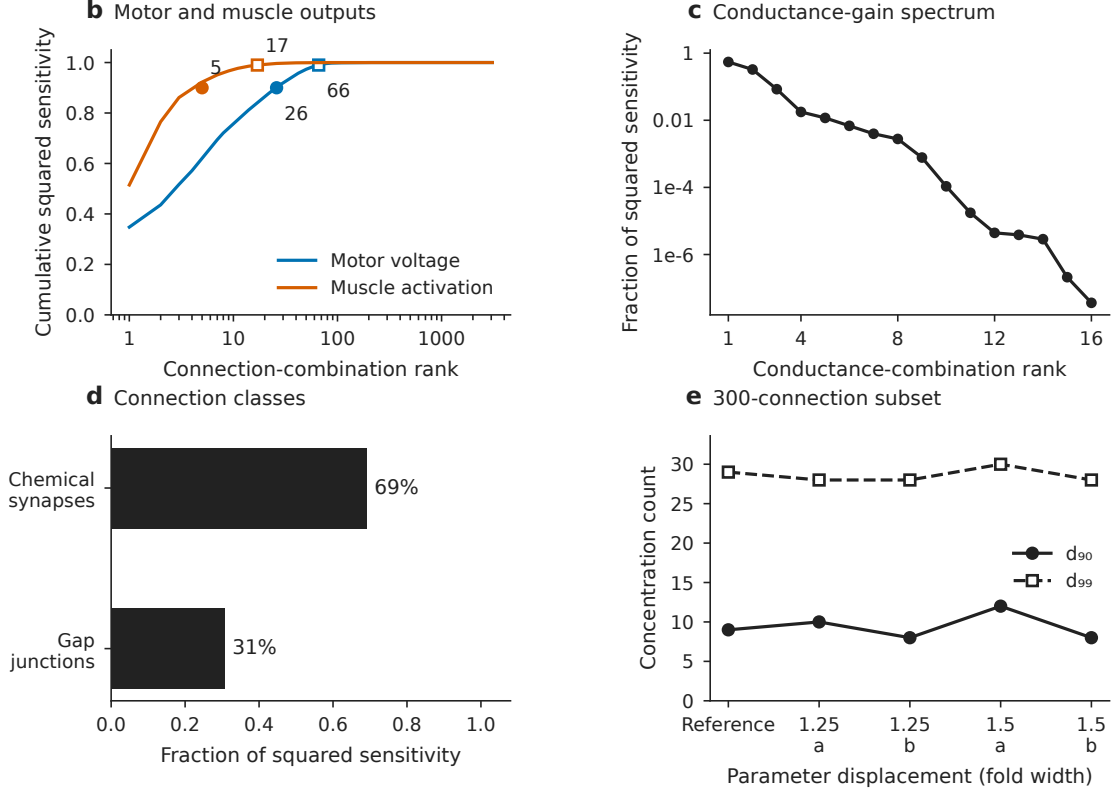
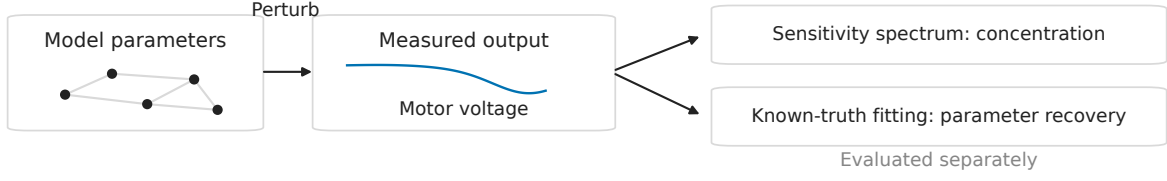


Figure 1: Measuring sensitivity concentration and parameter recovery. **a**, Overview of sensitivity analysis and known-truth fitting. Arrows show analysis flow; the network is schematic and the blue trace shows recorded baseline motor voltage after the initial transient. **b**, Cumulative spectral mass from all 3076 eigenvalues for motor voltage (blue) and the model’s clipped muscle activation (vermillion), computed from the same perturbation rollouts. Filled circles mark d_{90} and open squares d_{99} ; labels give their counts. The horizontal position is eigenvalue rank on a logarithmic axis. **c**, Fraction of total squared sensitivity per eigenvalue for the sixteen conductance-class gains. The vertical axis is logarithmic. **d**, Contributions of chemical-synapse and gap-junction weights to the trace of the connection sensitivity matrix. Bars and percentages show the same quantity. **e**, d_{90} (filled circles, solid line) and d_{99} (open squares, dashed line) at the reference point and four multiplicatively displaced parameter points in a separate nearby-point assay, measured over the same 300 connections. For each weight, a displacement factor is sampled independently and uniformly on a logarithmic scale between the reciprocal of the fold width and the fold width. Labels a and b denote independent displacements. Black denotes condition-neutral quantities.

Dependence on the output and perturbation scale

Reducing the output vector can hide sensitivity directions before any inference is attempted. A Jacobian with n output components has rank at most n , regardless of the number of parameters. Conversely, many time samples need not supply independent information. We examined this dependence using the same 48 actuator gains of the FlyGym walking controller (23). At a relative perturbation step of 0.25, ten summary statistics give $d_{90} = 1$ and $d_{99} = 1$, whereas 4200 joint-angle samples give 11 and 31 (Fig. 2a). This comparison changes the recorded output while keeping the parameter set and step fixed.

Perturbation size also changes the measured spectrum. Across relative steps of 0.05, 0.15, 0.25 and 0.50, the joint-angle response divided by the step is 1.98, 1.39, 1.34 and 1.31. The corresponding d_{90} values are 16, 12, 11 and 11 (Fig. 2b,c). The smallest step emphasizes realization variability (Supplementary §2); the two largest steps give similar response-to-step ratios and spectral concentration. At 0.50, some simulations diverge and two gains have no usable perturbation pair, so we use 0.25 for the readout comparison.

The same full-connectome perturbations also change their apparent concentration across the model’s output layers. The linear projection of motor voltages into muscle outputs, before clipping, gives concentration counts 3/9, while the model’s clipped activation gives 5/17, compared with motor-voltage 26/66 (Fig. 1b). Each readout uses its own fixed baseline temporal scales. Thus the output layer and its normalization alter the measured concentration, even with the connection perturbations held fixed.

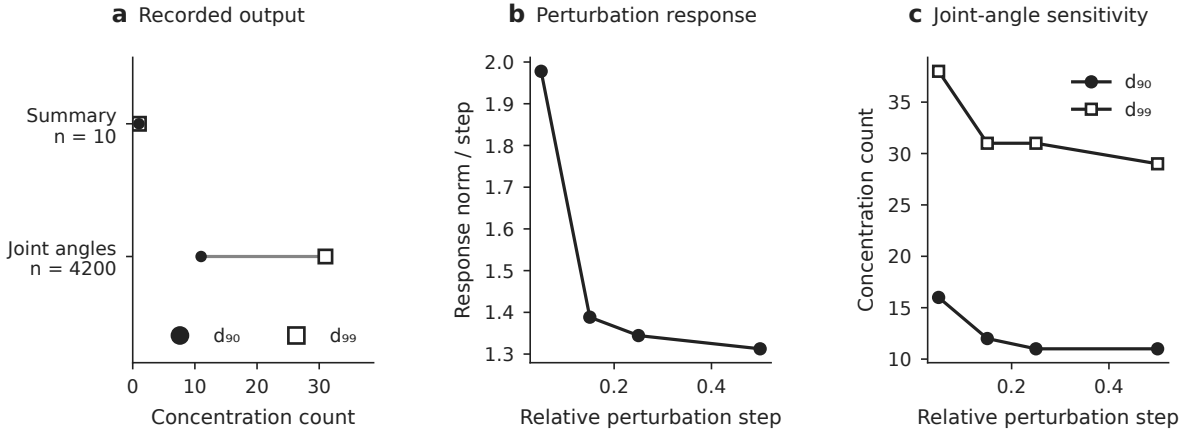


Figure 2: **Recorded output and perturbation scale change the measured concentration.** **a**, d_{90} (filled circles) and d_{99} (open squares) for ten summary statistics and 4200 joint-angle samples from the same 48 FlyGym gains at step 0.25. Connecting segments link the two concentration thresholds. **b**, Median standardized response norm divided by the relative step. **c**, Joint-angle d_{90} (filled circles) and d_{99} (open squares) across the same steps. Lines connect measured settings. The step-0.50 estimate omits two gains without usable simulations. Five controller seeds are averaged within each perturbation measurement; markers are point estimates and carry no uncertainty interval.

Mechanisms contributing to motor-output sensitivity

We next replaced individual connection weights with sixteen named conductance gains, each applied uniformly to cells expressing the corresponding mechanism. The resulting coordinates represent mechanism classes, rather than independent per-cell conductances. Their finite-step motor-voltage response has $d_{90} = 3$ and $d_{99} = 6$, and all sixteen gains produce a measurable response. The inward rectifier IRK, sodium leak NCA and potassium conductances EGL-36 and EGL-2 have the largest column norms. EGL-19 and UNC-2 are relatively weak under this input and voltage readout (Supplementary §3).

The eigenvectors distinguish class sensitivity from combinations of classes. For a normalized eigenvector, participation is the inverse sum of fourth powers of its loadings: it describes the number of coordinates contributing appreciably to that direction. The leading direction has participation 2.88 across the sixteen coordinates, with NCA providing the largest loading; the weakest direction has participation 1.02 and is dominated by SLO-1/EGL-19. Thus, strongly sensitive combinations involve a few conductance classes, while some weak combinations are close to single classes.

We also measured sensitivity at nearby parameter points. Using an identical 300-connection subset at the reference point and four displaced points, d_{90} ranges from 8 to 12 and d_{99} from 28 to 30. Leading subspaces have mean pairwise principal-angle cosine 0.89, compared with 0.16 in a matched random-subspace control. Concentration persists across these sampled points (Fig. 1e). Its rollout and output weighting differ from those of the full-connectome measurement.

Visual-network state and sensitivity concentration

The connectome-constrained flyvis network provides a sensory-circuit comparison (25). We load one pretrained network and compare it with the library’s default initialized state. Its 734 parameters comprise 65 biases, 65 time constants and 604 synaptic strengths. Positive time constants and strengths use log coordinates, while biases use a fixed affine scale. The trained network has 31 zero-strength boundary coordinates. We exclude these zero-strength connections from both states, leaving the same 703 parameter coordinates; output weights are fixed to the trained baseline.

We average activity over the final five stimulus frames and group contiguous neuron indices into bins. With 64 neural-activity bins, the trained network gives $d_{90} = 9$ and $d_{99} = 22$, compared with 15 and 34 in the initialized network (Fig. 3a). All 64 readout directions are numerically resolved under the stated tolerance, illustrating the difference between rank and concentration. Reducing the recorded bins changes both counts (Fig. 3b). Both states receive the same sinusoidal luminance movie and use the same output metric.

Finite-difference and integration-step controls preserve strong concentration while exposing discrepancies in weak responses (Fig. 3c; Supplementary §4). The implementation’s time-constant floor changes the effective dynamics when the integration step is reduced.

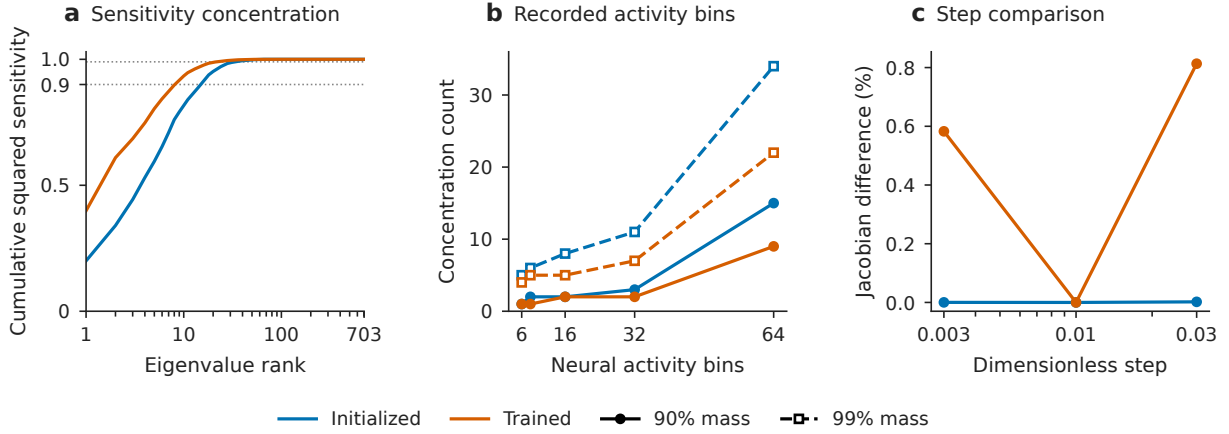


Figure 3: **Visual-network sensitivity depends on model state and recorded output.** Blue denotes the default initialized reference; vermilion denotes one pretrained network. Both use the same 703 parameters after excluding trained zero-strength connections, with output scales fixed from the trained baseline. **a**, Cumulative mass of the complete sensitivity spectrum; dotted grey guides mark 90% and 99%. **b**, Concentration counts at each measured number of neural-activity bins. Filled circles with solid lines give the 90% count; open squares with dashed lines give the 99% count. **c**, Relative Frobenius-norm difference of the standardized Jacobian from the step-0.01 estimate. Curves join measured settings; no sampling intervals are estimated. Individual weak-response diagnostics are given in Supplementary §4.

Stimulation conditions and combined sensitivity

Changing what a circuit is driven to do can change which perturbations affect its output. In a closed-loop BAAIWorm assay recording path statistics, chemosensory gain perturbations produce standardized output changes of 0.872 with food close and 0.122 with food far, a 7.15-fold contrast. Motor gain perturbations show the opposite ordering, 0.109 and 0.357 (Fig. 4a).

Combining stimulation conditions exposes additional numerically resolved sensitivity directions. For 20 gains, the two conditions have 4 and 4 resolved sensitivity directions, respectively, at relative singular-value tolerance 10^{-6} , while the stacked assay has 8 (Fig. 4b). Its concentration counts are 2/4 for chemotaxis, 1/2 for locomotion and 2/4 for the combined assay. The 14 neuron-class leak gains similarly give 4/4/8 resolved directions, with combined concentration 1/4. For the twenty-gain comparison, the combined assay records more resolved sensitivity directions even though its concentration counts equal those of chemotaxis.

Within the FlyGym walking controller, sensitivity remains concentrated across the four descending commands tested. Averaging over all subsets of a given size, d_{90} increases from 12.25 for individual commands to 13.00 for all four; d_{99} increases from 30.75 to 32.00 (Fig. 4c). These combined conditions change the concentration counts little.

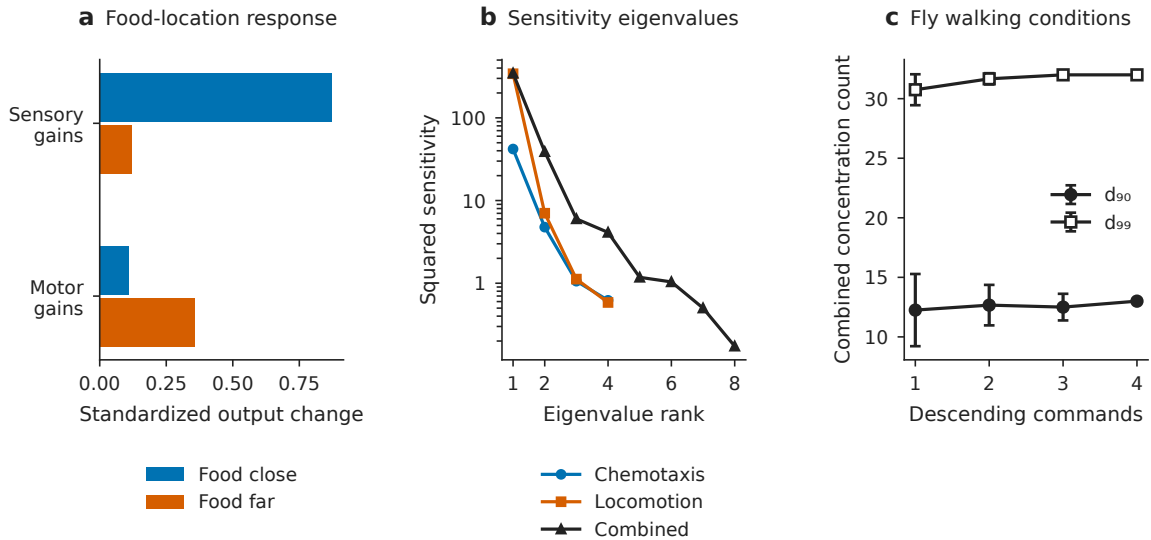


Figure 4: **Context changes sensitivity without necessarily increasing its concentration count.** **a**, Standardized changes in path observables after chemosensory and motor gain perturbations, with food close (blue) and far (vermillion). **b**, Eigenvalues of the complete twenty-gain chemotaxis (blue circles), locomotion (vermillion squares) and combined assay (black triangles). The combined matrix is the sum of the two condition matrices with their stated fixed normalization. Lines connect eigenvalue ranks. Very small eigenvalues are omitted from the logarithmic display; concentration counts use the complete spectra. **c**, d_{90} (filled circles) and d_{99} (open squares) of the summed FlyGym sensitivity matrices across four descending commands at step 0.25. Markers give means over all subsets of the indicated size; error bars give the standard deviation across subsets. Black denotes the combined measurement.

Prospective stimulus comparison

We next tested whether a set of models fitted to a baseline observation could guide the choice of an additional stimulus and improve prediction. In 12 independent simulated parameter cases, each strategy observed one noisy baseline and one noisy additional voltage trajectory. The strategy selects a stimulus using a Gaussian score of predicted response variation across baseline-fitted models relative to observation noise. Controls repeat the baseline, select a random candidate or select the largest-response candidate. Mean voltage root-mean-square error (RMSE) on the excluded stimulus was 0.115 mV for ensemble selection, compared with 0.053, 0.049 and 0.039 mV for the three controls. All paired descriptive bootstrap intervals for the mean differences crossed zero. This experiment does not demonstrate an average predictive improvement from the selection rule (Supplementary §8).

Sensitivity concentration and parameter recovery

Cell models let us compare a sensitivity measurement with recovery of known generating parameters. We examined eleven models spanning neuronal excitability and biochemical oscillations. Starts are random multiplicative displacements around the generating parameter vector, with fold

widths 1.5, 3 and 10. For each model, we report both the fraction of fits whose output error falls below 0.02 and the achieved parameter error among those fits (Fig. 5b,d). Parameter error is a normalized Euclidean distance in the model’s stated units.

Successful output fits give accurate parameter recovery in some models and substantial error in others. In the Hodgkin–Huxley panel, all 72 starts meet the output threshold, with median normalized parameter error 1.85×10^{-5} . Morris–Lecar and FitzHugh–Nagumo also give small errors among successful fits. In contrast, MAPK reaches the output threshold in 5 of 24 starts, yet its median parameter error among those fits is 0.40.

The Hodgkin–Huxley assay grid separates concentration from achieved recovery more directly. We vary the free parameter count (3, 5 or 8) and record either one spike-count statistic, seven current–frequency and spike-shape features, or voltage traces with 200 or 1000 samples. The measured d_{90} is 1 in all twelve combinations. For the seven-feature assay, median parameter error rises from 3.8% with three free parameters to 38% with eight, while median output error remains below 0.02. In the trace-based combinations, the median fit fails the output threshold, so its parameter error also reflects optimization failure (Fig. 5c). A single concentration count therefore cannot specify how many parameters a fitting procedure will recover.

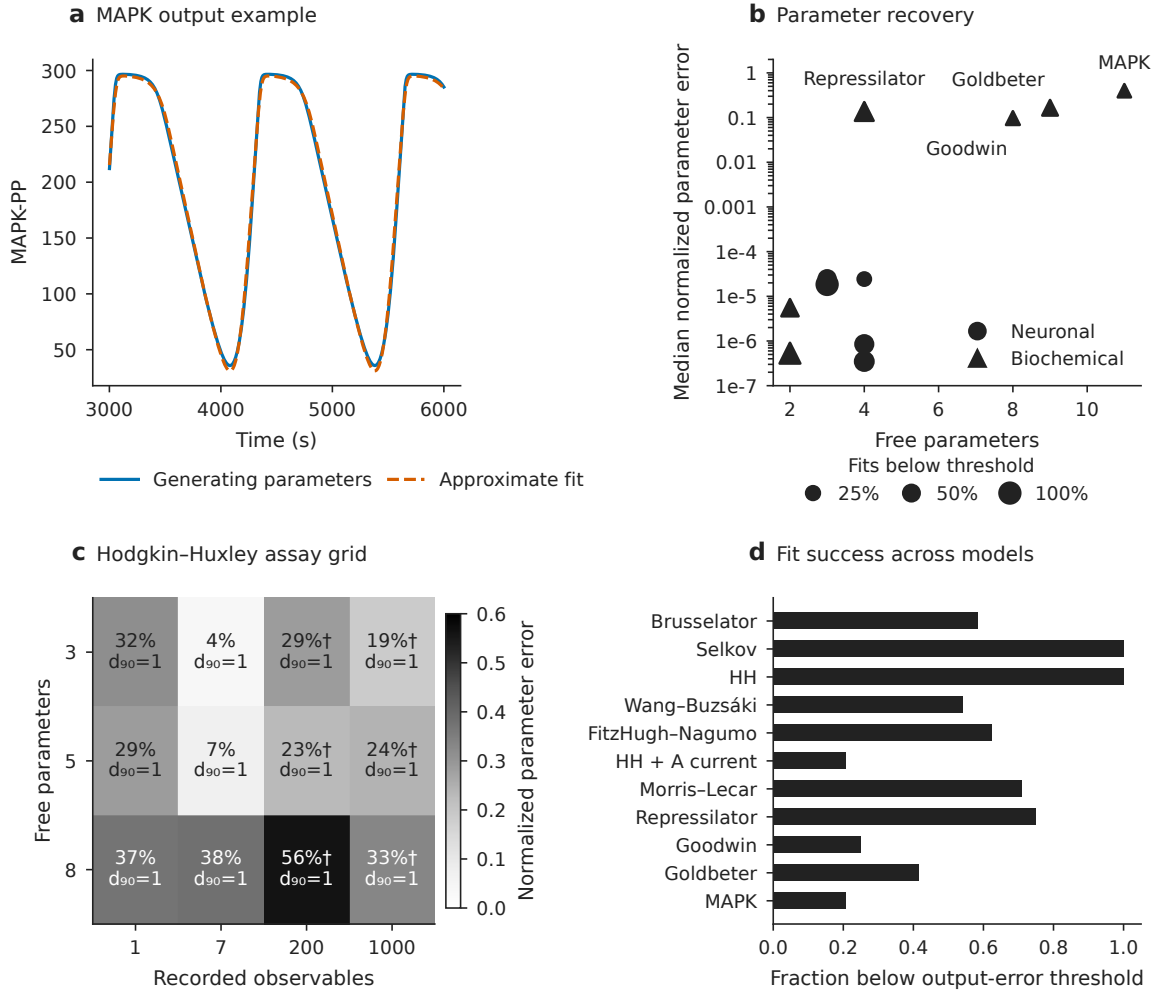


Figure 5: Concentrated sensitivity and accurate parameter recovery are separate properties. **a**, Doubly phosphorylated MAPK (MAPK-PP) at the generating parameters (blue, solid) and at an approximate fitted set (vermillion, dashed). The displayed fit has normalized parameter error 0.43 and feature loss 0.028, above the 0.02 fit threshold; it illustrates trace similarity and is excluded from the successful-fit summary. **b**, Median normalized L_2 parameter error among fits below the output threshold, against free parameter count. Circles denote neuronal models and triangles biochemical models. Marker size indicates the fraction of successful fits, as illustrated by the size key. **c**, Hodgkin–Huxley assay grid. Cell shading and percentages give median normalized L_2 parameter error; printed d_{90} values give the concentration count. Daggers mark a median output error at or above 0.02. Axes specify parameter and observable counts. **d**, Fraction of fits below the output threshold for every model in **b**. Parameter-error normalization and fitting protocols are model-specific.

Discussion

Output sensitivity is strongly concentrated in the connectome-scale nematode circuit under the measured input and readout. Individual weight perturbations change motor-neuron voltage, yet a small set of combinations carries most of the squared response. Comparisons of conductance gains and connection classes locate the components contributing to that response. Neural voltages, muscle readouts, joint angles and path statistics expose different aspects of a model, and reducing those outputs can reduce the span of sensitivity before parameter fitting begins. The spectrum depends on parameter coordinates and output normalization as well as the model and readout.

A concentration count describes relative sensitivity, whereas parameter recovery also depends on its magnitude and the precision of observations. Multiplying every eigenvalue by the same positive factor leaves the concentration counts unchanged, even though all output responses become stronger. Weak directions can still be estimated from sufficiently precise data. Conversely, a poor fit can leave parameters inaccurate even when the observation map contains information about them. The cell-model experiments connect these distinctions to achieved parameter accuracy, extending work on sloppiness and inference (6, 11, 13).

The span of resolved sensitivity directions can expand while d_{90} remains nearly unchanged, because additional directions may carry little weight compared with the leading one. The complete twenty-gain nematode spectra provide a direct example: the combined assay resolves more directions than either condition alone while retaining the chemotaxis concentration counts. The FlyGym command set likewise retains concentrated combined sensitivity. These findings motivate examining both the directions supplied by a new assay and their magnitudes. Iterative experimental design has improved parameter estimation in biochemical models (26–28). Ensemble selection did not demonstrate an average predictive improvement in our simulated comparison, which used an observation-noise model and an excluded prediction condition. The value of a new measurement therefore also depends on the model ensemble, fitting procedure and measurement cost.

Biological interpretation requires relating model parameters to experimental perturbations. Class-level sensitivity rankings concern uniform model gains; gene-level perturbations can change multiple mechanisms and cells. Exploratory comparisons with two fly connectomes and a nematode signal-propagation atlas did not support transfer of the sensitivity rankings to those animal measurements (Supplementary §6). Postural variation, electrophysiological heterogeneity and sequence conservation provide separate biological context, with distinct observables and levels of description. The numerical interpretation is likewise specific to the measured perturbation scale: some weak, oscillator and locomotor responses do not admit a validated local derivative.

Methods

Model systems and measurements

The primary system is the point-neuron circuit derived from BAAIWorm (22), simulated with NEURON (17). Its 136 cells and 3076 connection weights are driven by fixed sensory input. The full-connectome assay records 80 motor-neuron voltages; closed-loop body trajectories are assessed in the separate context experiments. Conductance assays perturb named gains uniformly across expressing cells. FlyGym (23) provides actuator-gain measurements from summary statistics and joint angles. The flyvis visual model (25) is described with its explicit network state and coordinates in Supplementary §4. Cell-model equations and fitting protocols are specified in Supplementary §5; other systems are assessed in Supplementary §2 and §7.

Parameter coordinates and sensitivity matrix

For nonzero connection weights and positive gains we use dimensionless relative coordinates, implemented either as a multiplicative central perturbation $\theta_i(1 \pm \delta)$ or as $\theta_i e^{\pm \delta}$. Signs of inhibitory weights are held fixed, so their local coordinates correspond to log magnitude. At zero displacement the two perturbation schemes have the same derivative; finite-step estimates need not coincide. Signed or zero-valued flyvis parameters use the separately specified affine or one-sided coordinates. The raw Jacobian J is standardized as $K = W^{1/2}J$ using fixed reference output scales. The sensitivity matrix is $H = K^\top K = J^\top W J$. Here, W encodes the fixed reference output scales.

For coordinates q , the sensitivity matrix is $H = J_q^\top W J_q$, with non-negative eigenvalues up to floating-point error. For a locally linear transformation $q = Tz$, $H_z = T^\top H_q T$, so concentration depends on parameter scaling. The concentration count is

$$d_\alpha = \min \left\{ k : \sum_{i=1}^k \lambda_i \geq \alpha \sum_i \lambda_i \right\},$$

where eigenvalues are ordered from largest to smallest. We report d_{90} and d_{99} , and the participation ratio $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$. Numerical rank is reported separately using an explicit tolerance on a complete spectrum. The number of output components bounds the number of sensitivity directions that an assay can record; other parameter directions can affect other readouts.

Numerical diagnostics and combined conditions

We compare repeat baselines, response norms divided by perturbation step, and derivative estimates across steps where available. Repeat baselines assess realization variability under identical settings, and step comparisons assess changes in the finite-difference estimates. A roughly constant response-to-step ratio supports a finite-step interpretation over that tested range. Discontinuous or phase-sensitive responses are described separately from validated differential measurements. For matched parameter coordinates, combined conditions use $H_{\text{combined}} = \sum_c H_c$ with the stated per-condition weights and fixed baseline output scales. The rank of this sum describes the span of the recorded directions; its concentration counts describe their relative contributions.

Known-truth parameter fitting

The eleven cell models include Hodgkin–Huxley, its A-current variant, Wang–Buzsáki, Morris–Lecar, FitzHugh–Nagumo, MAPK, repressilator, Goodwin, Selkov, Brusselator and Goldbeter oscillators (29–39). Starts satisfy $\theta_i^{\text{init}} = \theta_i^* e^{\eta_i}$, with independent $\eta_i \sim \mathcal{U}(-\ln f, \ln f)$ and $f \in \{1.5, 3, 10\}$. Thus, random displacements are centered on the known truth. Features are computed on settled traces at full resolution and then reduced. The fitting loss combines amplitude quantiles, Fourier magnitudes and autocorrelation features, normalized by the mean absolute target-feature value. Fits below loss 0.02 supply the primary conditional recovery summaries. Parameter error is $\|\hat{\theta} - \theta^*\|_2 / \|\theta^*\|_2$ in the model’s stated units; fit fractions are reported alongside it. The assay grid additionally reports medians across all starts, with unsuccessful output matching marked explicitly.

Statistics and biological context

Displaced-point subspaces are compared through principal-angle cosines and a matched random-orientation control. FlyGym command curves average the summed sensitivity over all subsets of each size; their error bars describe subset variability. The cell panel reports descriptive conditional

medians and fit fractions; cross-model parameter-count comparisons are descriptive. The external comparisons in Supplementary §6 are exploratory and retain their sampling, coverage and dependence limitations. Model-population PCA, animal-feature PCA and sensitivity spectra are treated as different quantities.

Use of AI tools

Generative AI assistants were used for language editing and grammar, and to assist with analysis code and figure assembly. The author designed the study, verified all analyses and outputs, and takes full responsibility for the content. No AI tool is an author.

Data availability

The study uses publicly available simulator inputs and biological reference datasets. Result files and analysis outputs accompany the source package; each supplementary section identifies the files supporting its measurements. The previously archived version is available at <https://doi.org/10.5281/zenodo.22120856> (40).

Code availability

Analysis code, figure scripts and checks of the stored spectra and numerical claims accompany the manuscript. The previously archived analysis repository remains available. Forward-model reproduction uses the specified simulator installations and inputs; the accompanying outputs also support recalculation of the reported spectra.

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Author contributions

C.H. conceived the study, designed and performed all analyses, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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